INCH-POUND

MIL-T-27/77A
4 April 1989
SUPERSEDING
MIL-T-27/77
29 August 1975

# MILITARY SPECIFICATION SHEET

## TRANSFORMER, POWER, TF5T03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.

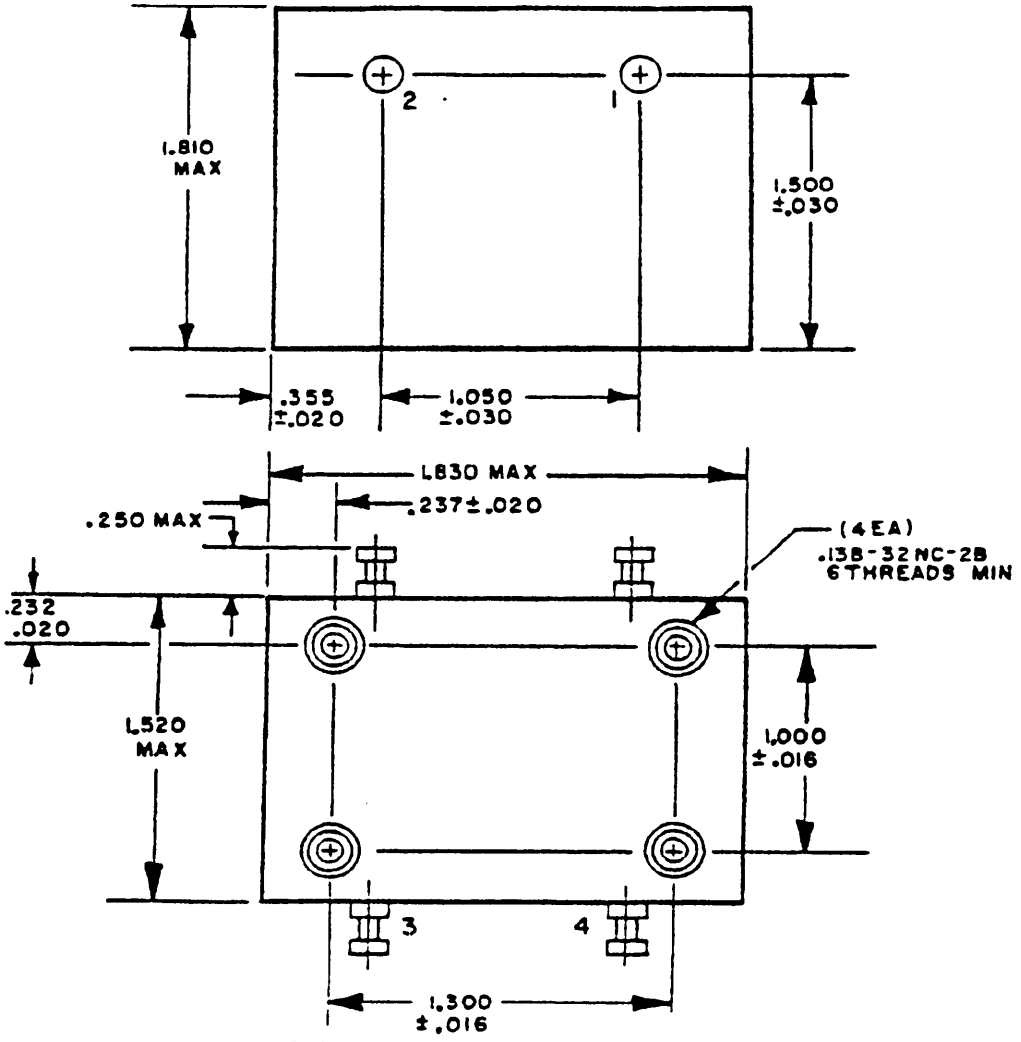


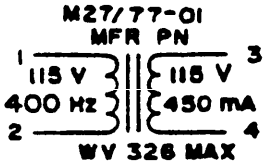

FIGURE 1. Dimensions and configuration.

Ⓐ denotes changes



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CIRCUIT DIAGRAM
AND MARKING

| Inches | mm | Inches | mm |
|---|---|---|---|
| .016 | 0.41 | .355 | 9.02 |
| .020 | 0.51 | 1.000 | 25.40 |
| .030 | 0.76 | 1.050 | 26.67 |
| .138 | 3.51 | 1.300 | 33.02 |
| .232 | 5.89 | 1.500 | 38.10 |
| .237 | 6.02 | 1.520 | 38.61 |
| .250 | 6.35 | 1.810 | 45.97 |
| | | 1.830 | 46.48 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary current: .495 ampere.

Volt-ampere: 56.9 volt-amperes

Primary voltage:

(1-2): 115 volts rms, 400 hertz.

Secondary voltage:

(3-4): 115 volts rms, 0.45 ampere rms.

Working voltage:

(1-2): 326 volts maximum.

(3-4): 326 volts maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Diallyl phthalate.

Terminals: Solder type, brass, tin plated.

Terminal height: 0.25 inch maximum.

Ⓐ Weight: 8.5 ounces maximum.

Operating temperature range: -55°C to +170°C.

Ⓐ Altitude: 10,000 feet maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 4.5 pounds.

Dielectric withstanding voltage (at sea level): 1,000 volts rms.

Electrical characteristics:

No load: With 115 volts, 400 hertz across (1-2), current in (1-2), shall not exceed 100 milliamperes and the power in (1-2) shall not exceed 5 watts.

Voltage across (3-4): 123 volts ±1 percent.

Rated load: With 115 volts, 400 hertz across (1-2), the voltage across (3-4) shall be 115 volts ±3 percent.



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DC resistance:

(1-2): 6.5 ohms ±20 percent.

(3-4): 8.7 ohms ±20 percent.

Temperature rise: +70°C maximum.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/77-01.

## CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 11, 17, 80
DLA - ES

User activities:
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0731-02)



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